



Arachis Online Training

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Practical Manufacturing Lessons

YOGHURT FORMULA

Practical Small-Scale Production Guide

Batch Size 10 Litres	Product Type Set or Drinking Yoghurt	Level Beginner Friendly
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Prepared by Arachis Online Training

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Important: This document is a practical training guide for small-scale production. Yoghurt is a food product, so hygiene, correct heating, clean packaging, and refrigeration are not optional.

1. Ingredients for 10 Litres

Ingredient	Quantity	Purpose
Fresh milk or full cream milk	10 litres	Main yoghurt base
Sugar	800 g - 1 kg	Sweetness
Skim milk powder or full cream milk powder	300 g - 500 g	Improves thickness and body
Plain live yoghurt or yoghurt starter culture	200 ml - 300 ml	Fermentation starter
Flavour	20 ml - 50 ml	Taste
Food colour	Small amount	Appearance
Potassium sorbate or sodium benzoate	5 g - 10 g	Helps extend shelf life when used with good hygiene and refrigeration
Clean water	As needed	For dissolving sugar or powder if necessary

2. Equipment Needed

- Stainless steel pot
- Stove or gas burner
- Thermometer
- Clean bucket with lid
- Whisk or clean mixing stick
- Measuring scale
- Measuring jug
- Clean cloth
- Clean bottles, cups, or sachets
- Cooler box or warm place for incubation

3. Step-by-Step Production Method

Step 1: Prepare and clean everything

Wash all equipment properly using hot water and detergent. Rinse well. If possible, sterilise bottles or cups with hot water. Cleanliness is very important because yoghurt spoils easily when contaminated.

Step 2: Heat the milk

Pour 10 litres of milk into a clean pot. Heat slowly to about 85°C while stirring to prevent burning. Hold the milk at 85°C for 10-15 minutes. This kills unwanted bacteria and improves thickness.

Step 3: Add milk powder and sugar

While the milk is still hot, add 300 g - 500 g milk powder and 800 g - 1 kg sugar. Stir until fully dissolved. Use 500 g milk powder for thicker yoghurt and 300 g for a lower-cost product.

Step 4: Cool the milk

Allow the mixture to cool to 42°C - 45°C. Do not add starter while the milk is too hot, because high heat can kill the yoghurt bacteria.

Step 5: Add the starter culture

At 42°C - 45°C, add 200 ml - 300 ml plain live yoghurt. Mix gently but thoroughly. Use plain yoghurt labelled live cultures or active cultures. Do not use flavoured yoghurt as a starter.

Step 6: Incubate the yoghurt

Cover the container and keep it warm for 6-8 hours. The best incubation temperature is around 40°C - 45°C. You can use a cooler box with warm water bottles, wrap the bucket with blankets, or keep it in a warm area away from direct fire.

Do not shake or disturb the yoghurt during incubation.

Step 7: Check if it has set

After 6-8 hours, check the yoghurt. It should be thick and have a slightly sour yoghurt smell. If it is still too runny, leave it for another 1-2 hours.

Step 8: Cool the yoghurt

Once it has set, put it in a fridge or very cool place. Cooling stops fermentation and improves texture. Cool for at least 2-4 hours.

Step 9: Add flavour, colour, and preservative

After cooling, stir gently until smooth. Add flavour, a small amount of food colour, and preservative. Mix well. Suitable flavours include strawberry, banana, vanilla, raspberry, peach, and mixed fruit.

Step 10: Package

Pack into clean bottles, cups, or sachets. Keep refrigerated. Do not expose finished yoghurt to heat or dirty handling.

4. Shelf Life and Quality Control

Without preservative, yoghurt may last about 5-7 days under refrigeration. With strong hygiene, refrigeration, and correct preservative use, it may last about 10-14 days. Always test your product before selling widely.

Do not sell yoghurt with a bad smell, gas bubbles, swollen bottles, mould, bitter taste, or slimy texture.

5. Low-Cost Drinking Yoghurt Version

This version is thinner and easier to drink. It is suitable where customers prefer bottled drinking yoghurt.

Ingredient	Quantity
Milk	10 litres
Sugar	800 g - 1 kg
Milk powder	200 g - 300 g
Plain live yoghurt starter	200 ml
Flavour	20 ml - 50 ml
Food colour	Small amount
Preservative	5 g - 10 g

6. Thick Premium Yoghurt Version

This version gives a richer, thicker product suitable for cups or premium bottles.

Ingredient	Quantity
Milk	10 litres
Sugar	1 kg
Milk powder	500 g - 700 g
Plain live yoghurt starter	300 ml
Flavour	30 ml - 50 ml
Food colour	Small amount
Preservative	5 g - 10 g

7. Common Problems and Solutions

Problem	Possible Causes	Solution
Yoghurt is too runny	Low incubation temperature; weak or dead starter; not enough milk powder; short incubation time	Use fresh live starter, keep warm at 40°C - 45°C, and increase milk powder.
Yoghurt is too sour	Incubated too long; temperature too high; not cooled quickly after setting	Reduce incubation time and refrigerate immediately after setting.
Water separates on top	Disturbing during incubation; weak milk solids; over-fermentation	Increase milk powder and avoid shaking while yoghurt sets.

Problem	Possible Causes	Solution
Spoils quickly	Dirty equipment; poor refrigeration; dirty bottles; poor preservative control	Improve hygiene, sterilise packaging, and keep yoghurt cold.

8. Production Notes for Business

- Use clean water, clean containers, clean hands, and clean packaging at every stage.
- Do not rely only on preservative. Preservative helps, but it cannot fix dirty production or poor storage.
- Keep batch records: date made, ingredient quantities, starter used, incubation time, and number of bottles produced.
- Start with small test batches before producing for many customers.
- For public sale, follow local food safety and labelling requirements where applicable.

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